

ILLUMINATION AND SHADING

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Illumination Theory

Illumination theory, also known as lighting theory or shading theory, is a fundamental concept in computer graphics and computer-generated imagery (CGI) that focuses on simulating and understanding the behavior of light and its interaction with objects in a virtual or rendered environment. Illumination theory aims to create realistic and visually appealing images by accurately modeling the way light interacts with surfaces and materials.

There are several components and models within illumination theory that contribute to the overall lighting and shading effects in computer graphics. Here are some key aspects of illumination theory

Light Sources:

Illumination theory considers the various sources of light in a scene, such as point lights, directional lights, spotlights, or even global illumination techniques like ambient light or indirect lighting. The properties of light sources, including intensity, color, and position, influence the overall illumination of the scene.

Surface Materials:

Different surface materials interact with light in distinct ways. Illumination theory takes into account material properties, such as diffuse reflectance (how much light is scattered in different directions), specular reflectance (how light is reflected in a specific direction, resulting in highlights), transparency, and more. Models like the Phong

reflection model or the more complex physically-based rendering (PBR) models attempt to simulate these material properties accurately.

Light Transport Models: Light transport models, such as the ray-tracing algorithm or rasterization with shading techniques, simulate how light rays interact with objects in a scene. These models calculate the interactions between light sources, surfaces, and observers (the virtual camera or viewer) to determine the color and intensity of light reaching each pixel in the rendered image.

Shading Models: Shading models define how light interacts with surfaces and determine the appearance of objects in the rendered scene. These models consider factors like diffuse shading (Lambertian reflection), specular shading (highlight reflection), ambient shading (indirect illumination from the environment), and more. Shading models provide mathematical formulas or algorithms to compute the final color and brightness of each pixel.

Global Illumination: Global illumination techniques aim to simulate the indirect lighting effects that occur in real-world scenes, including inter-reflections, caustics, and color bleeding. Advanced algorithms like radiosity, photon mapping, or path tracing are employed to capture the complex behavior of light bouncing and interacting with surfaces.

Illumination and Surface Rendering:

- Realistic displays of a scene are obtained by perspective projections and applying natural lighting effects to the visible surfaces of object.
- An illumination model is also called lighting model and sometimes called as a shading model which is used to calculate the intensity of light that we should see at a given point on the surface of an object.
- A surface-rendering algorithm uses the intensity calculations from an illumination model.

Light source

In computer graphics, light sources are essential components used to simulate the emission of light within a virtual scene. They play a vital role in illuminating objects and surfaces, creating shadows, and determining the overall visual appearance of the rendered image. Light sources provide the necessary information for shading algorithms to compute the interaction between light and objects in the scene.

Light sources in computer graphics can be conceptualized as virtual objects that emit or radiate light energy. They are positioned, oriented, and characterized by various properties to achieve the desired lighting effects. Total reflected light from an opaque luminous object is the sum of

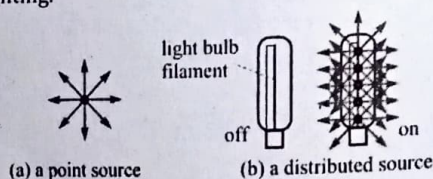
the contributions from the light source and other reflecting surfaces in the scenes. So a surface that is not directly exposed to a light source may still be visible if nearby objects are illuminated. A luminous object, in general, can be both a light source and a light reflector eg: plastic glove with a light bulb.

Point Light Source:

- Emits light uniformly in all directions from a single point in 3D space.
- Represents a localized light source, such as a light bulb or a candle flame.
- Light rays radiate equally in all directions from the point of emission.
- The intensity of light diminishes with distance from the point light source.
- Produces sharp, well-defined shadows in the presence of occluding objects.
- Commonly used to simulate small, concentrated light sources in computer graphics.

Distributed Light Source:

- Represents a light source that has a finite size and emits light uniformly across its surface.
- Can have various shapes, such as rectangular, spherical, or custom geometries.
- Emits light in a way that simulates soft shadows and diffused lighting.
- Light rays emanate from multiple points on the surface of the distributed light source.
- Provides a more natural and even illumination compared to a point light source.
- Commonly used for simulating larger light sources like area lights or sky lighting.

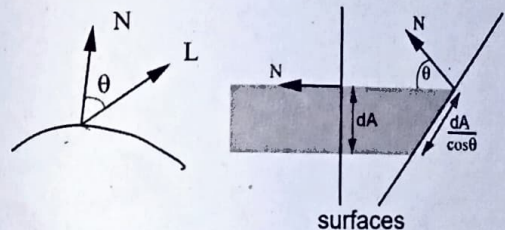


Light Reflection:

Light reflection refers to the phenomenon where light rays bounce off a surface after striking it. Reflection plays a significant role in how we perceive the appearance of objects and surfaces in the real world and is an important aspect of rendering in computer graphics. There are two primary types of light reflection:

Diffuse Reflection:

- Diffuse reflection is the scattering of light in various directions when it interacts with a rough or matte surface.
- When light hits a surface with microscopic irregularities, it bounces off in multiple directions.
- Diffuse reflection occurs uniformly across the surface, regardless of the viewer's position or angle of observation.
- It creates a soft and even illumination on the surface, without distinct highlights or reflections.
- Diffuse reflection is responsible for the perception of an object's base color or the overall brightness of a surface.
- This type of reflection is common on materials like paper, cloth, or unpolished surfaces.



Diffuse reflections are constant over each surface in a scene independent of viewing direction

k_d or diffuse reflection coefficient or diffuse reflectivity (0 to 1)

k_d is nearly 1 for highly reflective surface and k_d is 0 where light absorbs (black surfaces)

Diffuse reflection intensity at any point on the surface as

$$I_{\text{ambdiff}} = k_d \cdot I_d$$

Where I_{ambdiff} is ambient light due to diffusion and I_d is light due to diffusion assuming diffuse reflections from the surface are scattered with equal intensity in all directions independent of the

Viewing direction (called "ideal diffuse reflectors") also called Lambertian reflectors and governed by Lambert's Cosine Law.

If "angle of incidence" between incoming light direction and surface normal is θ

$$I_{\text{Diff}} = k_d \cdot I_L \cos \theta$$

where I_{Diff} is light due to diffusion

If N is unit normal vector to a surface and L is unit direction vector to the point light source then

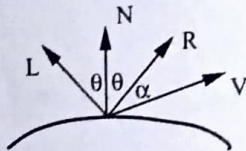
$$I_{\text{Diff}} = k_d \cdot I_L (N \cdot L)$$

In addition many graphics packages introduce an ambient reflection coefficient k_a to modify ambient light intensity k_a

$$\text{then } I_{\text{Diff (i.e. total)}} = k_a \cdot I_a + k_d \cdot I_L (N \cdot L)$$

1. Specular Reflection:

- Specular reflection is the mirror-like reflection of light from a smooth and polished surface.
- When light hits a surface, it reflects predominantly in a single direction according to the law of reflection.
- The angle of incidence (incoming light) is equal to the angle of reflection (outgoing light) relative to the surface normal.
- Specular reflection creates distinct highlights and shiny or glossy appearances on surfaces.
- The intensity and position of the specular highlights change with the viewer's position and the surface's orientation.
- This type of reflection is commonly observed on polished metals, glass, or other highly reflective surfaces.



$I = I_a k_a + I_d k_d (N \cdot L) + I_s k_s (R \cdot V)^n$ where n makes the specular highlight rapidly fall off

$$I = I_a k_a + I_d k_d (N \cdot L) + I_s k_s (R \cdot V)^n$$

This is the Phong shading model!!

Illumination model

An illumination model, also known as a lighting model or shading model, is a mathematical model used in computer graphics to simulate the interaction of light with objects in a virtual scene. It defines how light sources, materials, and surfaces affect the visual appearance of a

rendered image. Illumination models play a crucial role in creating realistic and visually appealing computer-generated imagery. There are various illumination models, each with its own set of equations and assumptions. However, most illumination models incorporate the following components:

- **Ambient Light:** Represents the overall indirect illumination in a scene that comes from all directions. It provides a base level of illumination and helps to avoid fully dark shadows.
- **Diffuse Reflection:** Simulates the scattering of light on rough or matte surfaces. It determines how much light is reflected from a surface based on the angle between the surface normal and the direction of the incident light.
- **Specular Reflection:** Models the mirror-like reflection of light on smooth or glossy surfaces. It produces shiny highlights and depends on the angle of incidence, the viewing direction, and the properties of the material.
- **Attenuation:** Describes how light intensity diminishes as it travels through space. It accounts for factors such as distance, absorption, and scattering.

Ambient light

Ambient light is a component of lighting in computer graphics used to simulate the overall indirect illumination present in a scene. Here's a short note on ambient light:

- Ambient light represents the scattered, reflected, and diffused light in a scene that comes from all directions uniformly.
- It provides a base level of illumination to all objects and surfaces, preventing fully dark shadows and creating a sense of overall brightness.
- Unlike other light sources, ambient light doesn't have a specific source location or direction associated with it.
- Ambient light contributes to the global illumination of a scene and helps establish the general lighting conditions.
- It is often used in conjunction with other light sources and shading techniques to create more realistic and visually appealing renderings.
- The intensity and color of ambient light can be adjusted to control the overall brightness and mood of the scene.
- While ambient light is essential for providing a starting point for illumination calculations, relying solely on ambient light can result

in a flat and unrealistic appearance. Hence, it is usually combined with other types of lighting to create more dynamic and believable lighting effects.

- In summary, ambient light is a foundational component of lighting in computer graphics that adds a base level of illumination to a scene, contributing to the overall visual quality and preventing fully dark areas.
- The equation expressing this simple model is

$$I = K_a$$

where I is the resulting intensity and K_a is the object's intrinsic intensity.

If we assume that ambient light impinges equally on all surface from all direction, then

$$I = I_a K_a$$

Where I_a is intensity of ambient light. The amount of light reflected from an object's surface is determined by K_a , the ambient-reflection coefficient.

K_a ranges from 0 to 1.

2. Diffuse reflection:

Objects illuminated by ambient light are uniformly illuminated across their surfaces even though light are more or less bright in direct proportion of ambient intensity. Illuminating object by a point light source, whose rays enumerate uniformly in all directions from a single point. The object's brightness varies from one part to another, depending on the direction of and distance to the light source.

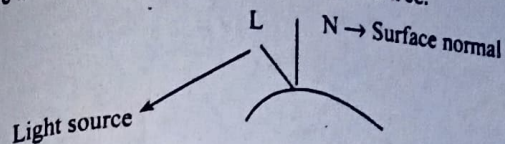
- The fractional amount of the incident light that is diffusely reflected can be set for each surface with parameter K_d , the coefficient of diffuse-reflection.
- Value of K_d is in interval 0 to 1. If surface is highly reflected, K_d is set to near 1. The surface that absorbs almost incident light, K_d is set to nearly 0.
- Diffuse reflection intensity at any point on the surface if exposed only to ambient light is

$$I_{\text{maddiff}} = I_a K_d$$

- Assuming diffuse reflections from the surface are scattered with equal intensity in all directions, independent of the viewing direction (surface called. "Ideal diffuse reflectors") also called Lambertian reflectors and governed by Lambert's cosine law.

$$I_{\text{diff}} = K_d I \cos \theta$$

where I is the intensity of the point light source.



If N is unit vector normal to the surface & L is unit vector in the direction to the point light source then

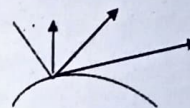
$$I_{\text{diff}} = K_d I (N \cdot L)$$

In addition, many graphics packages introduce an ambient reflection coefficient K_a to modify the ambient-light intensity I_a .

$$I_{\text{diff}} = K_a I_a + K_d I (N \cdot L)$$

3. Specular reflection and pong model

When we look at an illuminated shiny surface, such as polished metal, a person's forehead, we see a highlight or bright spot, at certain viewing direction. Such phenomenon is called specular reflection. It is the result of total or near total reflection of the incident light in a concentrated region around the "specular reflection angle = angle of incidence".



Let SR angle = angle of incidence as in figure.

N - unit vector normal to surface at incidence point

R - unit vector in the direction of ideal specular reflection

L - unit vector directed towards point light source.

V - unit vector pointing to the viewer from surface.

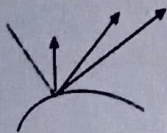
ϕ - the viewing angle relative to the specular reflection direction.

For ideal reflector (perfect mirror), incident light is reflected only in the specular reflection direction.

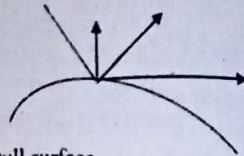
$\therefore V$ & R coincide ($\phi = 0$)

- Shiny surface have narrow ϕ and dull surface wider ϕ .
- An empirical model for calculating specular-reflection range developed by Phong Bui Tuong-called "Phong specular reflection model (or simply Phong model)", sets the intensity of specular reflection proportional to $\cos^n \phi \rightarrow 0$ to 90° .
- Specular reflection parameter n_s is determined by type of surface.

- Very shiny surface with large value n_s (say 100 or more) and dull surface, smaller n_s (down to 1).
- Rough surface such as chalk, $n_s = 1$.



Shiny surface
(large n_s)



Dull surface
(small n_s)

- Intensity of specular reflection depends upon material properties of the surface and θ . Other factors such as the polarization and color of the incident light.
- For monochromatic specular intensity variations can be approximated by SR coefficient $w(\theta)$.

Fresnel's law of reflection describes specular reflection intensity with θ and using $w(\theta)$, Phong specular reflection model as

$$I_{\text{spec}} = w(\theta) I_i \cos^n \phi$$

Where I_i is intensity of light source. ϕ is viewing angle relative to SR direction R .

For a glass, we can replace $w(\theta)$ with constant K_s specular reflection coefficient.

$$\text{So, } I_{\text{spec}} = K_s I_i \cos^n \phi$$

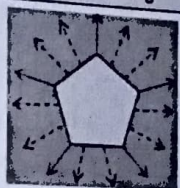
$$= K_s I_i (V \cdot R)^n$$

$$\text{Since } \cos \phi = V \cdot R$$

Diffuse vs specular Reflection

	Diffuse reflection	Specular reflection
Nature	Rough or matte surfaces	Smooth or glossy surfaces
Behavior	Light scatters in all directions	Light reflects in a concentrated direction
Surface normal	Incident angle = reflected angle	Incident angle = reflected angle
Intensity	Light intensity is spread evenly	Light intensity is concentrated in specific directions
Visibility	Visible from a wide range of angles	Visible only from specific angles

	Diffuse reflection	Specular reflection
Light source	Diffuse reflection does not create a clear reflection of the light source	Specular reflection creates a clear reflection of the light source
Examples	Paper, fabric, non-polished wood	Mirror, polished metal, glass

Flat shading	Gouraud shading	Phong shading
		
Simple and fast Phong equation only once per face	Still fast Phong equation at each vertex No 0th-order color discontinuities	Crisp highlights with few vertices
Mach bands	Slight mach bands, color invariance with quadrilaterals, problems with highlights	Slow Phong calculation for every pixel

Intensity Attenuation (IA)

Intensity is attenuated by the factor $\frac{1}{d^2}$ (d -distance that light has traveled)

Graphics packages have a general inverse quadratic attenuation function

$$F(d) = \frac{1}{a_0 + a_1 d + a_2 d^2}$$

User can fiddle with a_0, a_1, a_2 to obtain a variety of lighting effects for a scene

Basic illumination model is modified as

$$I = k_a I_a + \sum_{i=1}^0 f(d_i) I_{L_i} [k_d (N \cdot L_i) + K_s (N \cdot H_i)^n]$$

Intensity attenuation, also known as light attenuation, refers to the decrease in light intensity as it propagates through space or interacts with a medium. In computer graphics and rendering, intensity attenuation is a crucial factor in simulating realistic lighting effects. It accounts for how light diminishes in strength with distance or other factors such as absorption and scattering.

Polygon (Surface) Rendering Methods

Surface rendering methods are techniques used in computer graphics to generate a visual representation of the surfaces of 3D objects. These methods determine how light interacts with the surfaces and how the surfaces appear in the rendered image. To find the total light intensity, it interpolates the small set of intensity of certain pixels, calculated by using an illumination model across the surface

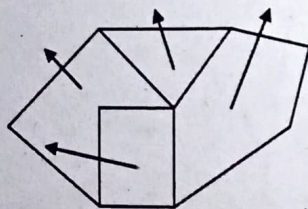
1. Constant Intensity Shading: (Flat Shading)

The simplest model for shading for a polygon is constant intensity shading also called as Faceted Shading or flat shading. This approach implies an illumination model once to determine a single intensity value that is then used to render an entire polygon. Constant shading is useful for quickly displaying the general appearance of a curved surface.

This approach is valid if several assumptions are true:

1. The light source is at infinity, so $N \cdot L$ is constant across the polygon face.
2. The viewer is at infinity, so $N \cdot V$ is constant across the polygon face.
3. The polygon represents the actual surface being modeled and is not an approximation to a curved surface.

Even if all conditions are not true, we can still reasonably approximate surface-lighting effects using small polygon facets with fast shading and calculate the intensity for each facet, at the centre of the polygon of course constant shading does not produce the variations in shade across the polygon that should occur.



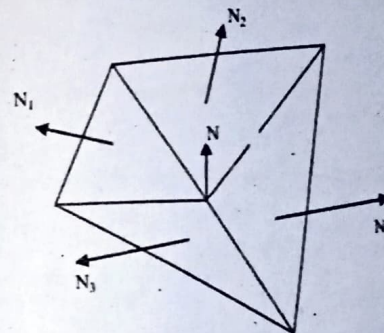
2. Interpolated shading

An alternative to evaluating the illumination equation at each point on the polygon, we can use the interpolated shading in which shading information is linearly interpolated across a triangle from the values determined for its vertices. Gouraud generalized this technique for arbitrary polygons. This is particularly easy for a scan line algorithm that already interpolates the z -value across a span from interpolated z -values computed for the span's endpoints.

Gouraud shading

Gouraud shading, also called intensity interpolating shading or color interpolating shading eliminates intensity discontinuities that occur in flat shading. Each polygon surface is rendered with Gouraud shading by performing following calculations.

1. Determine the average unit normal vector at each vertex. At each polygon vertex, we obtain a normal vertex by averaging the surface normals of all polygons sharing the vertex as:



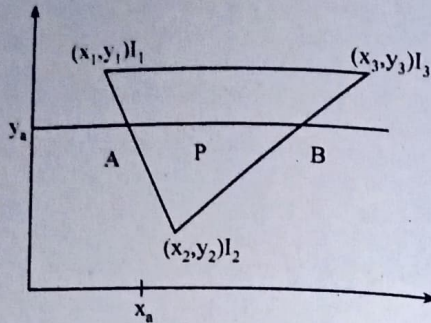
$$N_v = \frac{\sum_{k=1}^n N_k}{\left| \sum_{k=1}^n N_k \right|}$$

$$\text{Here, } N_v = \frac{N_1 + N_2 + N_3 + N_4}{|N_1 + N_2 + N_3 + N_4|}$$

where N_v is normal vector at a vertex sharing four surfaces as in figure.

2. Apply illumination model to calculate each vertex intensity.
3. Linearly interpolate the vertex intensity over the surface of the polygon.

Once N_v is known, intensity at the vertices can obtain from lighting model.



- Here in figure, the intensity of vertices 1,2,3 and I_1, I_2, I_3 are obtained by averaging normals of each surface sharing the vertices and applying a illumination model.
- For each scan line, intensity at intersection of line with polygon edge are linearly interpolated from the intensities at the edge end point.

So intensity at intersection point A, I_a is obtained by linearly interpolating intensities of I_1 and I_2 as

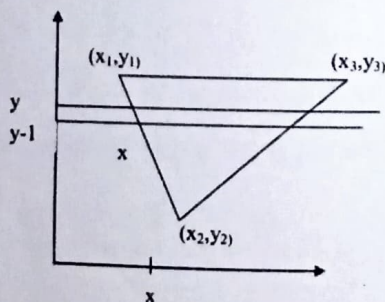
$$I_a = \frac{y_a - y_2}{y_1 - y_2} I_1 + \frac{y_1 - y_a}{y_1 - y_2} I_2$$

Similarly, the intensity at point B is obtained by linearly interpolating intensities at I_2 and I_3 as

$$I_b = \frac{y_3 - y_2}{y_3 - y_2} I_2 + \frac{y_3 - y_a}{y_3 - y_2} I_3$$

The intensity of a point P in the polygon surface along scan-line is obtained by linearly interpolating intensities at I_a and I_b as,

$$I_p = \frac{x_p - x_a}{x_b - x_a} I_b + \frac{x_b - x_p}{x_b - x_a} I_a$$



Then incremental calculations are used to obtain successive edge intensity values between scan-lines as:

$$I = \frac{y - y_2}{y_1 - y_2} I_1 + \frac{y_1 - y}{y_1 - y_2} I_2$$

Then we can obtain the intensity along this edge for next scan line at $y-1$ position as

$$I' = \frac{y-1 - y_2}{y_1 - y_2} I_1 + \frac{y_1 - (y-1)}{y_1 - y_2} I_2$$

$$= I + \frac{I_2 - I_1}{y_1 - y_2}$$

Similar calculations are made to obtain intensity successive horizontal pixel.

Advantages:

Removes intensity discontinuities at the edge as compared to constant shading.

Disadvantages:

Highlights on the surface are sometimes displayed with anomalous shape and linear intensity interpolation can cause bright or dark intensity streak called mach-bands.

Phong shading:

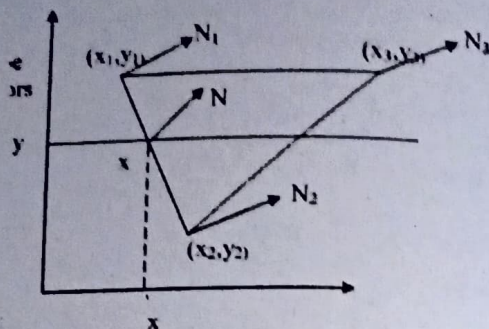
A more accurate method for rendering a polygon surface is to interpolate normal vector and then apply illumination model to each surface point. This method is called Phong shading or normal vector interpolation method for shading. It displays more realistic highlights and greatly reduce the mach band effect.

A polygon surface is rendered with Phong shading by carrying out following calculations.

- Determine the average normal unit vectors at each polygon vertex.
- Linearly interpolate vertex normals over the surface of polygon.
- Apply illumination model along each scan-line to calculate the pixel intensities for the surface point.

In figure, N_1, N_2, N_3 are the normal unit vectors at each vertex of polygon surface. For scan-line that intersect an edge, the normal vector N can be obtained by vertically interpolating normal vectors of the vertex on that edge as.

$$N = \frac{y - y_2}{y_1 - y_2} N_1 + \frac{y_1 - y}{y_1 - y_2} N_2$$



Incremental calculations are used to evaluate normals between scanlines and along each individual scan line as in Gouraud shading. Phong shading produces accurate results than the direct interpolation but it requires considerably more calculations.

Fast phong shading:

Fast Phong shading approximates the intensity calculations using a Taylor series expansion and Triangular surface patches. Since Phong shading interpolates normal vectors from vertex normals, we can express the surface normal N at any point (x, y) over a triangle as

$$N = A_x + B_y + C$$

Where A, B, C are determined from the three vertex equations.

$$N_k = Ax_k + By_k + C, k = 1, 2, 3 \text{ for } (x_k, y_k) \text{ vertex.}$$

Omitting the reflectivity and attenuation parameters

$$I_{diff}(x, y) = \frac{L \cdot N}{|L| \cdot |N|} = \frac{L \cdot (A_x + B_y + C)}{|L| \cdot |A_x + B_y + C|} = \frac{(L \cdot A)_x + (L \cdot B)_y + (L \cdot C)}{|L| \cdot |A_x + B_y + C|}$$

Re writing this

$$I_{diff}(x, y) = \frac{a_x + b_y + c}{(dx^2 + exy + fy^2 + gx + hy + i)^{1/2}} \dots \dots \dots (1)$$

Where $a, b, c, d, \dots$ are used to represent the various dot products as

$$a = \frac{L \cdot N}{|L|} \dots \dots \dots \text{and so on}$$

Finally, denominator of equation (1) can express as Taylor series expansions and relations terms up to second degree in x, y . This yields,

$$I_{diff}(x, y) = T_5x^2 + T_4xy + T_3y^2 + T_2x + T_1y + T_0$$

Where each T_k is a function of parameters $a, b, c, d, \dots$ and so forth.

This method still takes twice as long as in Gouraud shading. Normal Phong shading takes six to seven times that of Gouraud shading.

Gouraud vs Phong vs Fast phong

Shading technique	Calculation method	Normal Interpolation	Specular highlights	Performance
Gouraud shading	Per-vertex shading, interpolates values across the polygon	Interpolates normals per vertex	Less pronounced and smoothed	Faster rendering
Phong shading	Per-pixel shading, computes lighting equation at each pixel	Interpolates normals per pixel	Sharp and accurate	Slower rendering compared to Gouraud shading
Fast Phong shading	Precomputes shading values, interpolates across the surface	Interpolates normals per pixel	Less accurate than Phong shading	Faster rendering than traditional Phong shading

Color model

- **Additive colour:** It is obtained by adding together lights of different colors. The absence of all colors is black while the presence of all colors is white. It works with emitted light.
- **Subtractive colour:** It is obtained by subtracting lights of different colors. The absence of all colors is white while the presence of all colors is black. It works with reflected light.

Colour representation:

There are basically three modes of representing color combinations on a display:

- **Palette:** An indexed array usually with 256 entries, each three bytes long, representing the amounts of R, G, and B to be used to represent the color at the pixel in question.
- **Hi-color:** Sometimes referred to as "thousands of colors", uses 16 bits to represent colors, thus allowing 64K colors to be displayed.
- **True-color:** Sometimes referred to as "million of colors", uses 24 bits (3 bytes) to represent colors, thus allowing 256*256*256 colors to be displayed.

Colour concepts

Some of the key terms used in terminology are:

- **Shades:** Adding a pigment of black colour to some other colour produces a darker shade.
- **Tints:** Obtained by adding a white pigment to the original colour, making it lighter.
- **Tones:** They are produced by adding both black and white colour pigments.

An artist creates a painting by mixing colour pigments with white and black pigments to form various shades, tints, and tones, thus, giving the object a more realistic look in the graphical appearance.

Chromatic and Achromatic Light

Achromatic light can visualize only two-colour combination that is of black and white, as in black and white television and display monitors. Its only attributes are quantity that its intensity, luminance and brightness. Unlike achromatic light, the chromatic light consists of coloured light and their combinations, which produce visual sensation. Its attributes are hue, saturation, lightness and brightness of different colours. It refers to the perceived intensity of a self-luminous light source, such as light bulb, sun, etc.

A color model, also known as a color space or color system, is a mathematical representation that defines how colors can be represented and manipulated in digital or physical systems. Color models provide a standardized way to describe and represent colors, facilitating color reproduction, image processing, and color communication.

RGB

- The RGB color model is an additive color model in which red (R), green (G), and blue (B) are combined in various intensities to create a wide range of colors. It is based on the principle that when light of these three primary colors is mixed together at full intensity, they produce white light. By controlling the intensity of each primary color, different colors can be generated.

Here are some key aspects of the RGB color model:

- **Additive Color Mixing:** In the RGB model, colors are created by adding different amounts of red, green, and blue light. When all three primary colors are combined at their maximum intensity, they produce white. Conversely, when no light is present in any of the channels, the result is black.
- **Color Channels:** The RGB color model utilizes three color channels, one for each primary color. Each channel represents the intensity of

its respective color and is typically represented by an 8-bit value ranging from 0 to 255. For example, (255, 0, 0) represents pure red, (0, 255, 0) represents pure green, and (0, 0, 255) represents pure blue.

- **Color Representation:** In the RGB model, a color is represented as a combination of the intensities of the three color channels. Each color can be specified using a triplet of values (R, G, B), where R represents the intensity of red, G represents the intensity of green, and B represents the intensity of blue.

- **Color Gamut:** The RGB color model has a specific color gamut, which refers to the range of colors that can be represented. The gamut of the RGB model depends on the specific display or device being used, as different devices may have variations in their color reproduction capabilities.

- **Digital Displays:** The RGB color model is commonly used in digital displays, such as computer monitors, televisions, and mobile screens. These displays emit red, green, and blue light to create colors, and the RGB values determine the intensity of each color for each pixel on the screen.

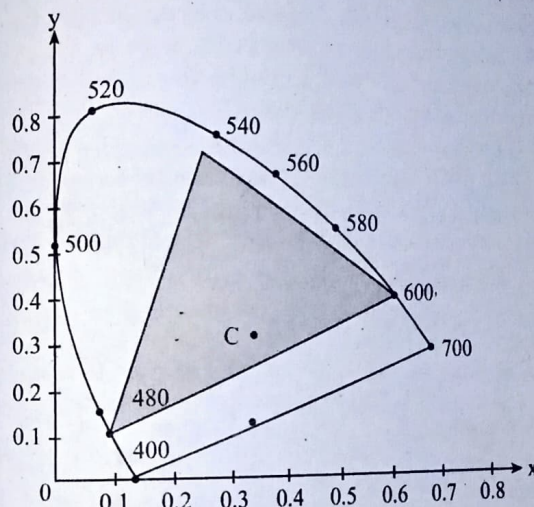
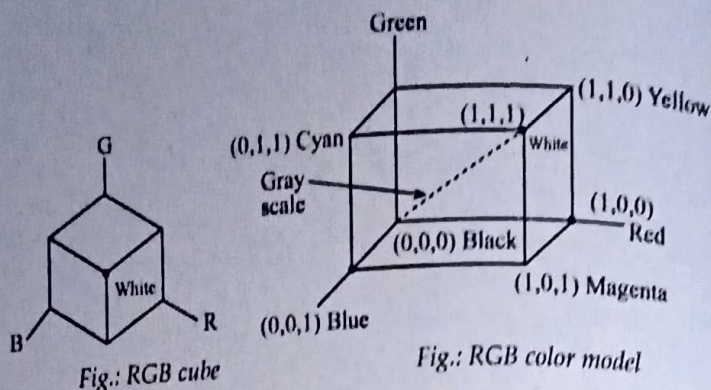


Fig.: RGB colour gamut

- **Image Editing and Processing:** RGB is a widely used color model in image editing and processing software. It allows for precise control over individual color channels, enabling adjustments such as color correction, color balance, and image enhancement.



CMY

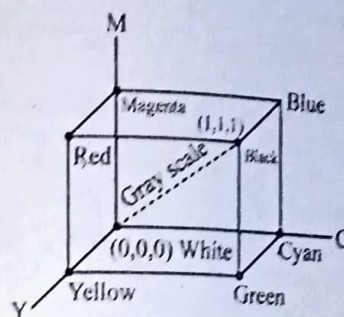
The CMY color model is a subtractive color model used in printing and color mixing. CMY stands for Cyan, Magenta, and Yellow, which are the primary colors in this model. In the CMY model, colors are created by subtracting specific wavelengths of light from white light, resulting in different colors. Here's how the CMY color model works:

- **Subtractive Color Mixing:** Unlike the additive color model (RGB), the CMY model works by subtracting color components from white light. When all three primary colors, cyan, magenta, and yellow, are combined at their maximum intensities, they absorb or subtract all wavelengths of light, resulting in black.
- **Primary Colors:** The CMY color model uses cyan, magenta, and yellow as the primary colors. Cyan absorbs red light, magenta absorbs green light, and yellow absorbs blue light. By subtracting these primary colors from white light, different colors are created.
- **Color Representation:** Colors in the CMY model are typically represented as a combination of the intensities of cyan, magenta, and yellow. Each color can be specified using a triplet of values (C, M, Y), where C represents the intensity of cyan, M represents the intensity of magenta, and Y represents the intensity of yellow. The intensity values usually range from 0 to 100%.
- **Color Mixing:** In the CMY model, colors are created by mixing different amounts of cyan, magenta, and yellow. When all three primary colors are combined at their maximum intensities (C = 100%, M = 100%, Y = 100%), they absorb all wavelengths of light, resulting in black. By reducing the intensity of each primary color, different colors can be achieved.
- **Color Gamut:** The CMY color model has a specific color gamut, which refers to the range of colors that can be represented. The

gamut of the CMY model is typically smaller than that of the RGB model, which means it cannot reproduce all the colors that can be represented in the RGB model.

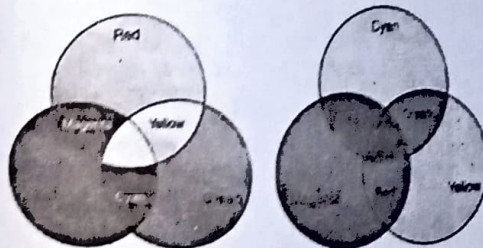
Printing: The CMY model is widely used in printing processes, where cyan, magenta, and yellow inks are combined in various amounts to produce a wide range of colors. By subtracting these colors from white paper, the desired colors are reflected back to the viewer.

Black Ink (K): In printing applications, a separate black ink component (K) is often added to improve color accuracy and conserve ink. This is known as the CMYK model, where K represents black. The addition of black ink helps create richer blacks and enhances the printing process.



RGB Colour Model
Additive colour model
For computer displays
Uses light to display color
Colors result from transmitted light
Red + Green + Blue = White

CMY Color Model
Subtractive color model
For printed materials
Uses ink to display color
Colors result from reflected light
Cyan + Magenta + Yellow = Black



CMYK model

The CMYK color model is a subtractive color model used primarily in printing processes. It is an extension of the CMY color model (Cyan, Magenta and Yellow) with the addition of a black component (K). The

CMYK model is designed to reproduce a wide range of colors accurately on printed media. Here's an overview of the CMYK color model:

Primary Colors:

- **Cyan (C):** Cyan is a bluish-green color that absorbs red light.
- **Magenta (M):** Magenta is a purplish-red color that absorbs green light.
- **Yellow (Y):** Yellow is a color that absorbs blue light.
- **Black (K):** The black component is added to the CMYK model to improve color accuracy, enhance the depth of black areas, and conserve ink. It is represented by the letter K to avoid confusion with the letter B, which represents blue in the RGB model.

Color Mixing:

- The CMYK model works by subtracting specific amounts of cyan, magenta, yellow, and black inks from white light to produce different colors.
- Mixing all four inks at maximum intensity should result in a very dark, nearly black color. By reducing the intensity of each ink, different shades, hues, and tones can be achieved.

Color Representation:

- Colors in the CMYK model are typically represented as a combination of the intensities of cyan, magenta, yellow, and black. Each color can be specified using a four-part value (C, M, Y, K), where C represents the intensity of cyan, M represents the intensity of magenta, Y represents the intensity of yellow, and K represents the intensity of black.
- The intensity values usually range from 0 to 100%, with 0% indicating no ink and 100% indicating maximum ink coverage.

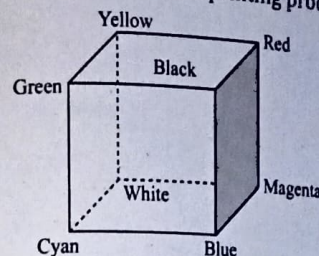
Color Gamut:

- The CMYK color model has a more limited color gamut compared to other color models, such as RGB. It cannot reproduce the full range of colors that can be achieved in RGB.
- Certain vibrant and highly saturated colors in the RGB color space may be challenging to reproduce accurately in CMYK, resulting in color shifts or less vibrant prints.

Printing Applications:

- The CMYK model is widely used in the printing industry for producing printed materials, such as magazines, brochures, posters, and packaging.
- Printers utilize cyan, magenta, yellow, and black inks to generate a wide spectrum of colors by mixing and overlapping the inks in different amounts and combinations.

The CMYK color model plays a crucial role in achieving accurate and high-quality color reproduction in printing processes.



Solved Numerical Problems

- Q. For the rectangle plane defined by points A(0,0), B(1,0), C(1,1), D(0,1), find the reflected intensity at point P(0.5,0.5) using Gouraud Shading technique. The average intensities of reflected illumination at the four vertices are: $I_A = 8$, $I_B = 9$, $I_C = 2$, $I_D = 4$.

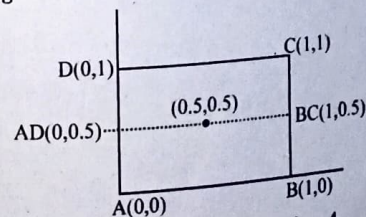
Solution:

Gouraud shading technique. The average intensities of reflected illumination at the four vertices are $I_A = 8$, $I_B = 9$, $I_C = 2$, $I_D = 4$.

1. Determine the average unit normal vector at each polygon vertex,
i.e. for any vertex position v , normal vector is given by

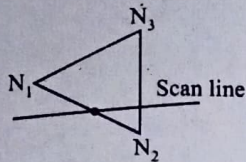
$$N_v = \frac{\sum_{k=1}^n N_k}{\left| \sum_{k=1}^n N_k \right|}$$

2. Apply an illumination model to each vertex to calculate the vertex intensity.
3. Linearity interpolate the vertices intensities over the surface of the polygon.



The average intensity between AD: $I_A = 8$, $I_D = 4$.

$$N = \frac{y - y_2}{y_1 - y_2} N_1 + \frac{y_1 - y}{y_1 - y_2} N_2$$



$$\begin{aligned} I_{AD} &= \frac{y_{AD} - y_D}{y_A - y_D} I_A + \frac{y_A - y_{AD}}{y_A - y_D} I_D \\ &= \frac{(0.5 - 1)}{(0 - 1)} (8) + \frac{0 - 0.5}{0 - 1} (4) \\ &= 6 \end{aligned}$$

Average intensity between BC:

$$\begin{aligned} I_{BC} &= \frac{y_{BC} - y_C}{y_B - y_C} I_B + \frac{y_B - y_{BC}}{y_B - y_C} I_C \\ &= \frac{(0.5 - 1)}{(0 - 1)} (9) + \frac{0 - 0.5}{0 - 1} (2) \\ &= 5.5 \end{aligned}$$

Now, the reflected intensity of rectangle at point P(0.5,0.5) is

$$\begin{aligned} I_P &= \frac{x_P - x_{AD}}{x_{BC} - x_{AD}} I_{BC} + \frac{x_{BC} - x_P}{x_{BC} - x_{AD}} I_{AD} \\ &= \frac{0.5 - 0}{1 - 0} (5.5) + \frac{1 - 0.5}{1 - 0} (6) \\ &= 5.75 \end{aligned}$$

- Q. Find out the total intensity at the centroid of a triangle defined by A(2,1,1), B(0,1,1), C(0,0,1). When illuminated by a point light source of intensity $I_l = 0.6$ at (2,2,6) using illumination model. The viewer is at (2,3,6). Assume ambient intensity $I_a = 0.1$ and parameters $K_a = 0.5$, $K_d = 0.8$, $K_s = 0.7$, take $n = 10$.

$$[\text{Centroid} = \frac{(x_1 + x_2 + x_3)}{3}, \frac{(y_1 + y_2 + y_3)}{3}, \frac{(z_1 + z_2 + z_3)}{3}]$$

[2079 Baishakh]

Solution:

The vertices of triangle A, B, C are

$$A = (2, 1, 1)$$

$$B = (0, 1, 1)$$

$$C = (0, 0, 1)$$

Position of light source = (2,2,6)

Position of the viewer = (2,3,6)

$$I_l = 0.6, I_a = 0.1$$

$$K_a = 0.5$$

$$K_d = 0.8$$

$$K_s = 0.7$$

$$n = 10$$

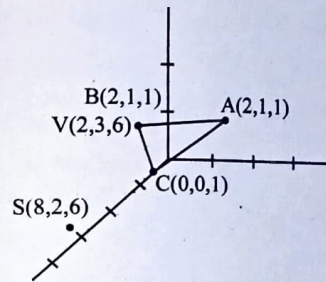
The centroid of the triangle is

$$A_c = \frac{2 + 0 + 0}{3} = 0.67$$

$$B_c = \frac{1 + 1 + 0}{3} = 0.67$$

$$C_c = \frac{1 + 1 + 1}{3} = 1$$

$$(A_c, B_c, C_c) = (0.67, 0.67, 1)$$



$$\text{Now, } I = I_a K_a + \frac{I_l K_d (\hat{N} \cdot \hat{L}) + I_l K_s (\hat{V} \cdot \hat{R})^{ns}}{K_c + K_d + K_s d^2}$$

Ambient light

$$I_a K_a = 0.1 \times 0.5 = 0.05$$

Diffuse light

$$\vec{N} = \vec{CA} \times \vec{CB}$$

$$= [(2-0)\vec{i} + (1-0)\vec{j} + (1-1)\vec{k}] \times [(0-0)\vec{i} + (1-0)\vec{j} + (1-1)\vec{k}]$$

$$= (2\vec{i} + \vec{j}) \times \vec{j}$$

$$= 2\vec{k}$$

$$\hat{N} = \vec{k}$$

$$\vec{L} = (2-0.67)\vec{i} + (2-0.67)\vec{j} + (6-1)\vec{k}$$

$$= 1.33\vec{i} + 1.33\vec{j} + 5\vec{k}$$

$$\hat{L} = \frac{1.33\vec{i} + 1.33\vec{j} + 5\vec{k}}{\sqrt{1.33^2 + 1.33^2 + 5^2}}$$

$$= 0.2489\vec{i} + 0.2489\vec{j} + 0.9358\vec{k}$$

$$I_{LK_d}(\hat{N} \cdot \hat{L}) = 0.6 \times 0.8 (\vec{k} \cdot (0.2489\vec{i} + 0.2489\vec{j} + 0.9358\vec{k}))$$

$$= 0.6 \times 0.8 \times 0.9358$$

$$= 0.44918$$

Specular light

$$\vec{R} = 2\hat{N}(\hat{N} \cdot \hat{L}) - \hat{L}$$

$$= 2 \times \vec{k}(\vec{k} \cdot (0.2489\vec{i} + 0.2489\vec{j} + 0.9358\vec{k}))$$

$$- (0.2489\vec{i} + 0.2489\vec{j} + 0.9358\vec{k})$$

$$= 2 \times \vec{k}(0.9358) - 0.2489\vec{i} + 0.2489\vec{j} + 0.9358\vec{k}$$

$$= -0.2489\vec{i} - 0.2489\vec{j} + 0.9358\vec{k}$$

$$\hat{R} = -0.249\vec{i} - 0.249\vec{j} + 0.9367\vec{k}$$

$$\vec{V} = (2, 3, 6) - (0.67, 0.67, 1)$$

$$= (1.33, 2.33, 5)$$

$$= 1.33\vec{i} - 2.33\vec{j} + 5\vec{k}$$

$$\hat{V} = 0.2344\vec{i} - 0.4106\vec{j} + 0.88117\vec{k}$$

$$\hat{R} \cdot \hat{V} = (-0.249\vec{i} - 0.249\vec{j} + 0.9367\vec{k}) \cdot (0.2344\vec{i} - 0.4106\vec{j} + 0.88117\vec{k})$$

$$= -0.05836 - 0.102239 + 0.82539$$

$$= 0.664791$$

$$I_{LK_s}(\hat{R} \cdot \hat{V}) = 0.6 \times 0.7 \times (0.664791)^{10} = 0.00708$$

$$\text{Total} = \text{Ambient} + \text{Diffuse} + \text{Specular}$$

$$= 0.05 + 0.44918 + 0.00708$$

$$= 0.50626$$

The total intensity at the centroid of a triangle is 0.50626 candela.